

6th Set of Problems

1. Consider the rotation matrix below, expressed in terms of a quaternion.

$$\mathbf{R}(\eta, \epsilon) = \begin{pmatrix} 2(\eta^2 + \epsilon_x^2) - 1 & 2(\epsilon_x \epsilon_y - \eta \epsilon_z) & 2(\epsilon_x \epsilon_z + \eta \epsilon_y) \\ 2(\epsilon_x \epsilon_y + \eta \epsilon_z) & 2(\eta^2 + \epsilon_y^2) - 1 & 2(\epsilon_y \epsilon_z - \eta \epsilon_x) \\ 2(\epsilon_x \epsilon_z - \eta \epsilon_y) & 2(\epsilon_y \epsilon_z + \eta \epsilon_x) & 2(\eta^2 + \epsilon_z^2) - 1 \end{pmatrix}$$

- Indicate which is the scalar and which is the vector part of the quaternion.
- Indicate the relation between the parameters of the vector-angle (angle and axis) representation and the quaternion parameters.
- Indicate the existing singularity in the vector-angle representation, and based on the previous question, show how it is eliminated by the quaternion representation.

2. For each shaded region in the block diagram for the *closed-loop inverse kinematics* algorithm, illustrated in Figure 1, gather the employed computations.

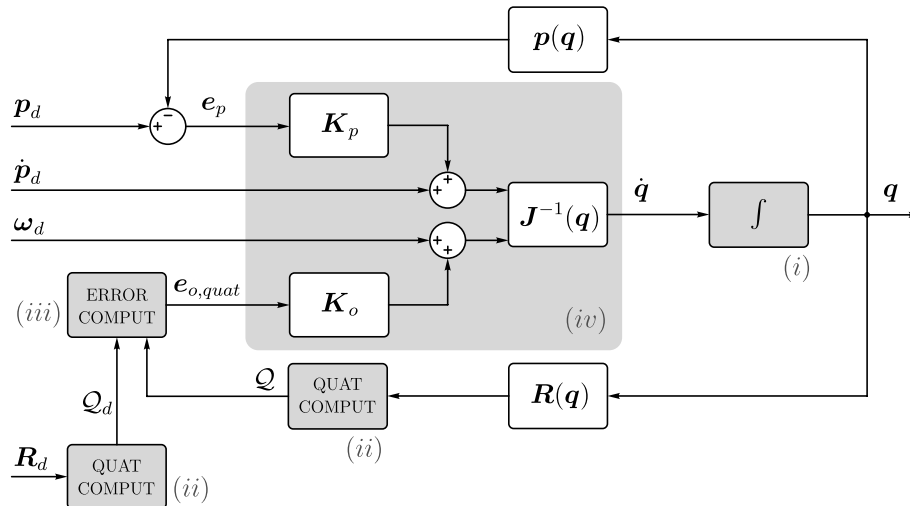


Figure 1: CLIK algorithm using the unit quaternion.

3. Consider the right hand of the DLR anthropomorphic manipulator, illustrated in Figure 2. Consider the index finger, comprised by the four revolute joints illustrated. The axes of joints one and two are orthogonal and do not intersect each other, being the distance between them ℓ_1 (not illustrated). The axes of joints two, three and four are parallel and the lengths of the respective links are ℓ_2 , ℓ_3 and ℓ_4 . Consider the end-effector as the finger's extremity.

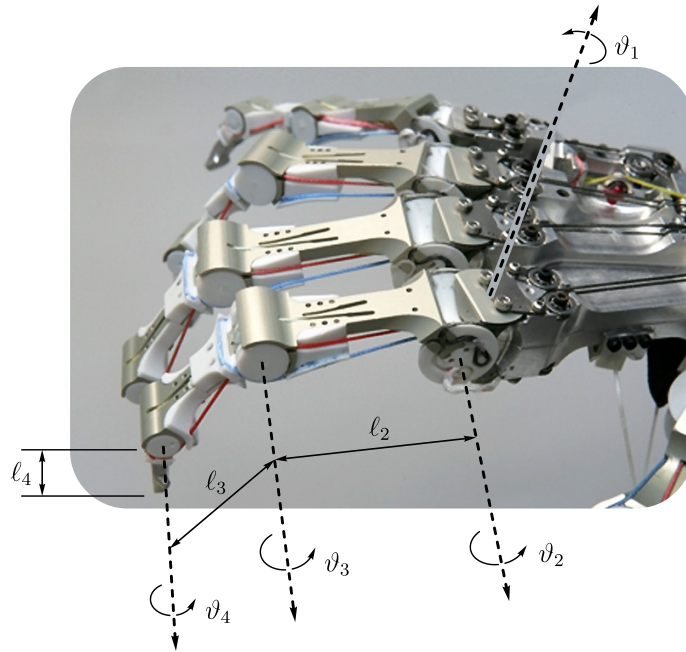


Figure 2: Robotic hand of the DLR *hand arm system*.

- (a) According to the Denavit-Hartenberg methodology, illustrate the reference frames over the robotic finger, and build the appropriate table of parameters. Indicate the configuration that you have chosen as *Home*.
- (b) Calculate the direct kinematics, $\mathbf{A}_4^o$.
- (c) Solve the corresponding inverse kinematics. Comment the solution, for the position and orientation of the end effector, in view of the four degrees of freedom of the finger.
- (d) Construct the Jacobian matrix for the robotic finger.
- (e) Show how the *closed-loop inverse kinematics*, in Problem 2, can be applied to the robotic finger.